

# Step 1: One-Sample Inference on Mean Log-Price

Danish Housing Prices · Mathematical Statistics

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2026-05-14

```
library(tidyverse)
dk_villa <- readRDS("../data/dk_villa.rds")
cat("n =", nrow(dk_villa), "\n")
```

n = 402

---

## 1 Statistical Model

We model the log-prices of Copenhagen Villas (2023–2024) as:

$$Y_i \sim \mathcal{N}(\mu, \sigma^2), \quad i = 1, \dots, n \quad (\text{independent})$$

With a single sample there is only one variance — no homogeneity test is applicable. The only distributional assumption to verify is normality.

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## 2 Normality Check

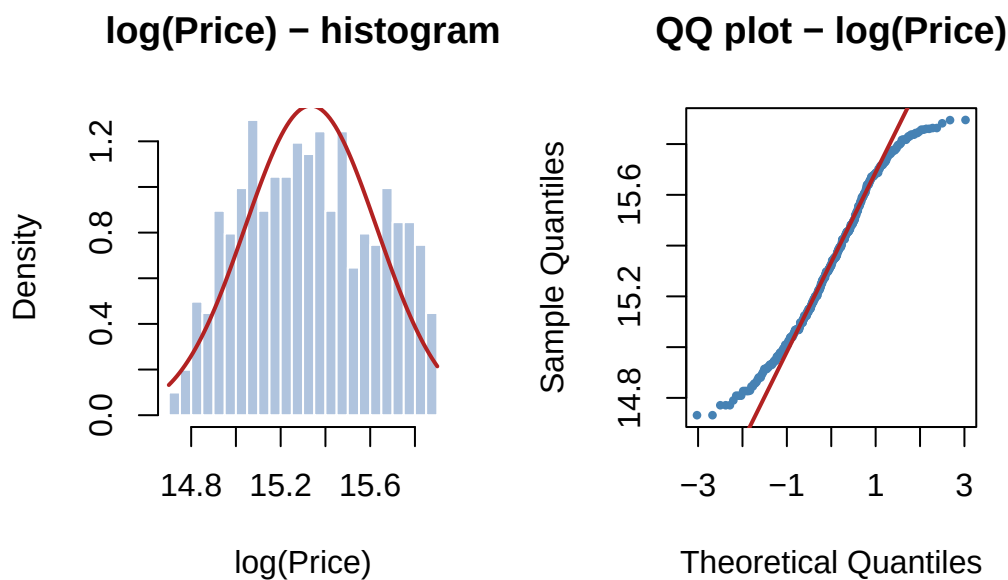
```

par(mfrow = c(1, 2))

hist(dk_villa$log_price,
     breaks = 30, prob = TRUE,
     main = "log(Price) - histogram",
     xlab = "log(Price)",
     col = "lightsteelblue", border = "white")
curve(dnorm(x, mean(dk_villa$log_price), sd(dk_villa$log_price)),
     add = TRUE, col = "firebrick", lwd = 2)

qqnorm(dk_villa$log_price,
     main = "QQ plot - log(Price)",
     pch = 16, cex = 0.6, col = "steelblue")
qqline(dk_villa$log_price, col = "firebrick", lwd = 2)

```



```

par(mfrow = c(1, 1))

```

### 3 Hypothesis Test

We test whether the mean log-price is consistent with a benchmark of 4,500,000 DKK:

$$H_0 : \mu = \log(4,500,000) \approx 15.32 \quad \text{vs} \quad H_1 : \mu \neq \log(4,500,000)$$

Under  $H_0$  the test statistic is:

$$T = \frac{\bar{Y} - \mu_0}{S/\sqrt{n}} \sim t(n-1)$$

```
mu0 <- log(4500000)
n <- nrow(dk_villa)
ybar <- mean(dk_villa$log_price)
s <- sd(dk_villa$log_price)
df <- n - 1

t_stat <- (ybar - mu0) / (s / sqrt(n))
p_val <- 2 * pt(-abs(t_stat), df = df)
ci <- ybar + c(-1, 1) * qt(0.975, df) * s / sqrt(n)

cat("n", n, "\n")
cat("y-bar", round(ybar, 4), "\n")
cat("s", round(s, 4), "\n")
cat("mu0", round(mu0, 4), "\n")
cat("t-statistic =", round(t_stat, 4), "on", df, "df\n")
cat("p-value", round(p_val, 6), "\n")
cat("95% CI for mu: [", round(ci, 4), "]\n")
cat("Back-transformed CI (DKK): [",
    format(round(exp(ci)), big.mark = ","), "]\n")
```

```
n          = 402
y-bar      = 15.3347
s          = 0.2941
mu0        = 15.3196
t-statistic = 1.0278 on 401 df
p-value    = 0.304677
95% CI for mu: [ 15.3058 15.3635 ]
Back-transformed CI (DKK): [ 4,438,492 4,702,038 ]
```

```
# Built-in verification
t.test(dk_villa$log_price, mu = mu0)
```

One Sample t-test

data: dk\_villa\$log\_price

```

t = 1.0278, df = 401, p-value = 0.3047
alternative hypothesis: true mean is not equal to 15.31959
95 percent confidence interval:
 15.30583 15.36351
sample estimates:
mean of x
 15.33467

```

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## 4 Confidence Interval for $\sigma$

A 95% CI for  $\sigma$  uses the fact that  $(n-1)S^2/\sigma^2 \sim \chi^2(n-1)$ :

$$\left[ \sqrt{\frac{(n-1)S^2}{\chi_{0.975}^2(n-1)}}, \sqrt{\frac{(n-1)S^2}{\chi_{0.025}^2(n-1)}} \right]$$

```

ci_sigma <- sqrt(c(
  (n - 1) * s^2 / qchisq(0.975, df),
  (n - 1) * s^2 / qchisq(0.025, df)
))
cat("95% CI for sigma: [", round(ci_sigma, 4), "]\n")

```

```
95% CI for sigma: [ 0.2751 0.316 ]
```

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## 5 Interpretation

**On the log scale:** the CI for  $\mu$  gives the range of plausible mean log-prices.

**On the DKK scale:** back-transforming via  $\exp(\cdot)$  yields a CI for the **median** price (not the mean), because for log-normal  $Y$ :  $\exp(\mathbb{E}[\log Y]) = \text{median}(Y)$ .

This is more informative for skewed price data, where the median is a more representative summary than the mean.